HAOYU HAN

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EDUCATION

Harvard University 09/2024–present
Ph.D. in Applied Mathematics
Working on optimization, reinforcement learning and semi-definite programming

University of Science and Technology of China (USTC) 09/2020–07/2024
B.S. in Information and Computational Science (Computational Mathematics)

EXPERIENCE

University of California, Berkeley 05/2026–08/2026
Visiting Student, hosted by Prof. Jason Lee; conducting research on optimization theory for machine learning.

Harvard University 07/2023–12/2023
Visiting Student, hosted by Prof. Heng Yang; conducted research on optimal control.

PUBLICATIONS

[1] **Haoyu Han**, Heng Yang. **On the Nonsmooth Geometry and Neural Approximation of the Optimal Value Function of Infinite-Horizon Pendulum Swing-up**. In: *Annual Learning for Dynamics and Control Conference (L4DC)*. 2024 (pdf)

[2] **Haoyu Han**, Heng Yang. **Building Rome with Convex Optimization**. In: *Robotics: Science and Systems (RSS)*. 2025 (web)

[3] Yulin Li, **Haoyu Han**, Shucheng Kang, Jun Ma, Heng Yang. **On the Surprising Robustness of Sequential Convex Optimization for Contact-Implicit Motion Planning**. In: *Robotics: Science and Systems (RSS)*. 2025 (web)

[4] Shucheng Kang, **Haoyu Han**, Antoine Groudiev, Heng Yang. **Factorization-free Orthogonal Projection onto the Positive Semidefinite Cone with Composite Polynomial Filtering**. Under Review: *SIAM Journal on Optimization*. 2025 (pdf)

[5] Alan Papalia, Nikolas Sanderson, **Haoyu Han**, Heng Yang, Hanumant Singh, Michael Everett. **Sparse Variable Projection in Robotic Perception: Exploiting Separable Structure for Efficient Nonlinear Optimization**. In: *IEEE International Conference on Robotics and Automation (ICRA)*. 2026 (pdf)

[6] **Haoyu Han**, Heng Yang. **Non-Uniform Noise-to-Signal Ratio in the REINFORCE Policy-Gradient Estimator**. In: *International Conference on Machine Learning (ICML)*. 2026 (pdf)

AWARDS

Best Systems Paper Award, Robotics: Science and Systems (RSS)	06/2025
China Optics Valley Scholarship (award students perform great in exams)	09/2023
National Scholarship (award students perform great in exams)	09/2022
Outstanding Student Scholarship	09/2021
Outstanding Freshman Scholarship	09/2020

RESEARCH EXPERIENCE

Computation and Analysis on Optimal Control of a Pendulum [1]

Keywords: Numerical ODE; Pontryagin's maximum principle; Value function — Harvard University

Individual Contributor *Advisor: Prof. Heng Yang* — 08/2023–12/2023

- Computed the cost-to-go function of the pendulum with error under 10^{-4}, with and without control constraints, using a novel contour-line methodology and Pontryagin's maximum principle (PMP).
- Discovered a non-smooth spiral in the cost-to-go function, calculated its geometry, and proved existence using symmetry and ODE theory.

Convex optimization for Structure-from-Motion (SfM) [2]

Keywords: SfM; Semidefinite Programming; Bundle adjustment — Harvard University

Individual Contributor *Advisor: Prof. Heng Yang* — 05/2024–12/2024

- Proposed a scaled bundle adjustment formulation that lifts 2D keypoint measurements to 3D with learned depth, solving it to global optimality with Semi-definite programming relaxations.
- Developed a CUDA solver dubbed XM and built an SfM pipeline with XM as the optimization engine; showed XM-SfM compares favorably in reconstruction quality while being significantly faster, more scalable, and initialization-free.

Non-uniform NSR for Policy-Gradient Estimators [6]

Keywords: Reinforcement Learning, Policy Gradient; Stochastic Gradient Descent — Harvard University

Individual Contributor *Advisor: Prof. Heng Yang* — 09/2025–present

- Developed a noise-to-signal ratio (NSR) perspective for the REINFORCE estimator, explaining late-stage slowdown/instability through NSR growth driven by shrinking policy covariance and near bang-bang action distributions.
- Derived exact finite-horizon NSR characterizations for (i) linear systems with Gaussian linear policies and (ii) polynomial systems with polynomial-Gaussian policies, enabling precise diagnostics of gradient estimator quality across parameters.

TEACHING

Teaching Assistant, Harvard ES/AM 158: Introduction to Optimal Control and Reinforcement Learning — 2025

SKILLS

Programming Languages	C++, Python, CUDA
Tools	LaTeX, Matlab, Mathematica, PyTorch, Markdown